

Supplementary Material

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1 More Details on Semi-supervised Learning

Semi-supervised Learning (SSL) aims to enhance model performance by using a small amount of labeled samples and a large amount of unlabeled samples. Compared with self-supervised learning, SSL combines labeled a large amount of unlabeled samples and unlabeled a large amount of unlabeled samples during the training process, which can effectively capture the relevant information of both. SSL can be roughly divided into three methods: One is the method based on consistency regularization which encourages models to produce consistent predictions when the same unlabeled sample is given different perturbations. The second is the self-training method. This method primarily use labeled samples to progressively generate pseudo-labels for unlabeled samples during training, treating these unlabeled samples as labeled samples to expand the training set. The last method combines consistent regularization and self-training methods.

2 More Details on Variational Auto-encoder

In the pre-training phase, we train a Variational Auto-encoder (VAE) [Kingma *et al.*, 2013] using unlabeled samples to learn a compact and meaningful latent representation of the input data. The loss function of the VAE, which is derived from variational inference principles, consists of two key components: the Kullback-Leibler (KL) divergence loss and the reconstruction loss. The overall loss function can be expressed as:

$$\begin{aligned} L_{vae} &= D_{KL} + L_{vr} \\ &= D_{KL} [q_\phi(z | U) || p(z)] - \mathbb{E}_{q_\phi(z|U)} [p_\theta(\bar{U} | z)] \\ &= \frac{1}{M} \sum_{j=N+1}^{N+M} \left[\frac{1}{2} \sum_{l=1}^L (\mu_{j,l}^2 + \sigma_{j,l}^2 - \log \sigma_{j,l}^2 - 1) \right. \\ &\quad \left. - \sum_{k=1}^K \left(u_{j,k} \cdot \log(\varphi_\theta(z_{j,k})) \right. \right. \\ &\quad \left. \left. + (1 - u_{j,k}) \cdot \log(1 - \varphi_\theta(z_{j,k})) \right) \right], \end{aligned} \quad (1)$$

where L and K , respectively, represent the dimension of latent representation and the dimension of input.

3 More Details on Pre-trained Auto-encoder

The loss in the pre-training stage is given by the sum of mask matrix reconstruction loss L_m , feature reconstruction loss L_f and contrastive loss L_{con} :

$$L_{pre} = L_f + \alpha L_m + \beta L_{con}, \quad (2)$$

where α and β are hyperparameters. The loss function L_m is formulated using the binary cross entropy:

$$\begin{aligned} L_m = -\frac{1}{M \cdot K} \left[\sum_{j=N+1}^{N+M} \sum_{k=1}^K (m_{j,k} \cdot \log f_\Delta(\tilde{z}_{j,k}) \right. \\ \left. - (1 - m_{j,k}) \cdot \log(1 - f_\Delta(\tilde{z}_{j,k})) \right]. \end{aligned} \quad (3)$$

It can be expressed L_f according to the mean square error loss:

$$L_f = \frac{1}{M \cdot K} \left[\sum_{j=N+1}^{N+M} \sum_{k=1}^K (u_{j,k} - f_\Delta(\tilde{z}_{j,k}))^2 \right], \quad (4)$$

where $\tilde{z}_{j,k} = g(f_\delta(\tilde{u}_{j,k}))$, $g(\cdot)$ is a projection head of two layers of neural networks, and activation functions are applied after all network layers. When the feature is categorical, the mean squared error is replaced by the cross entropy.

4 More Details on clinical laboratory test indicators

We used two publicly available datasets to evaluate the effectiveness of our method: the Adult Income dataset for binary classification, and the MNIST dataset for multi-class classification. The Adult Income dataset contains 32,561 instances with 104 feature dimensions and is a binary classification dataset for predicting whether an individual's income > \$50K. The MNIST dataset consists of 70,000 grayscale images of handwritten digits ranging from 0 to 9. Each image has a resolution of 28×28 pixels. We flatten each original 2D image into a 784-dimensional feature vector (28×28 = 784). The classification task on MNIST is to accurately predict the corresponding digit label (0-9) based on the 784-dimensional input vector, making it a multi-class classification task. Meanwhile, to determine factors associated with

irAEs, this study retrospectively collected clinical laboratory test data from 237 cancer patients who had received ICIs treatment and 4,817 cancer patients without irAEs labels from partner hospitals, including 87 indicators. After removing features with a large number of missing values, 41 features were obtained. The abbreviation reference table for the all collected indicators is presented in Table 1.

5 Visualization of SHAP values

We classified the collected clinical laboratory test data using Tab-semiSL. We then used the interpretability algorithm SHapley Additive exPlanations (SHAP) to rank the feature importance of the self encoder-predictor combined model after training. SHAP is a game-theoretic interpretability algorithm that determines the marginal contribution of each feature by computing its shapley value when added to a combination of other features [Shapley and others, 1953]. This value fairly attributes the contribution of each feature to the model output. For a given model, the shapley value of the feature k of the sample x_i is expressed as:

$$\text{Shapley}_k(x_i) = \sum_{S \subseteq K \setminus \{k\}} \frac{|S|!(|K| - |S| - 1)!}{|K|!} \times \left[\mathcal{P} \left(f_\delta \left(x_i^{(S \cup \{k\})} \right) \right) - \mathcal{P} \left(f_\delta \left(x_i^{(S)} \right) \right) \right] \quad (5)$$

where K represents the set of all the features of x_i , $S \subseteq K$, $x_i^{(S)}$ denotes the x_i with S selected features. We ranked the features according to shapley values and visualized the top-10 features contributing most to the model output in Figure 1.

References

- [Kingma *et al.*, 2013] Diederik P Kingma, Max Welling, et al. Auto-encoding variational bayes, 2013.
- [Shapley and others, 1953] Lloyd S Shapley et al. A value for n-person games. 1953.

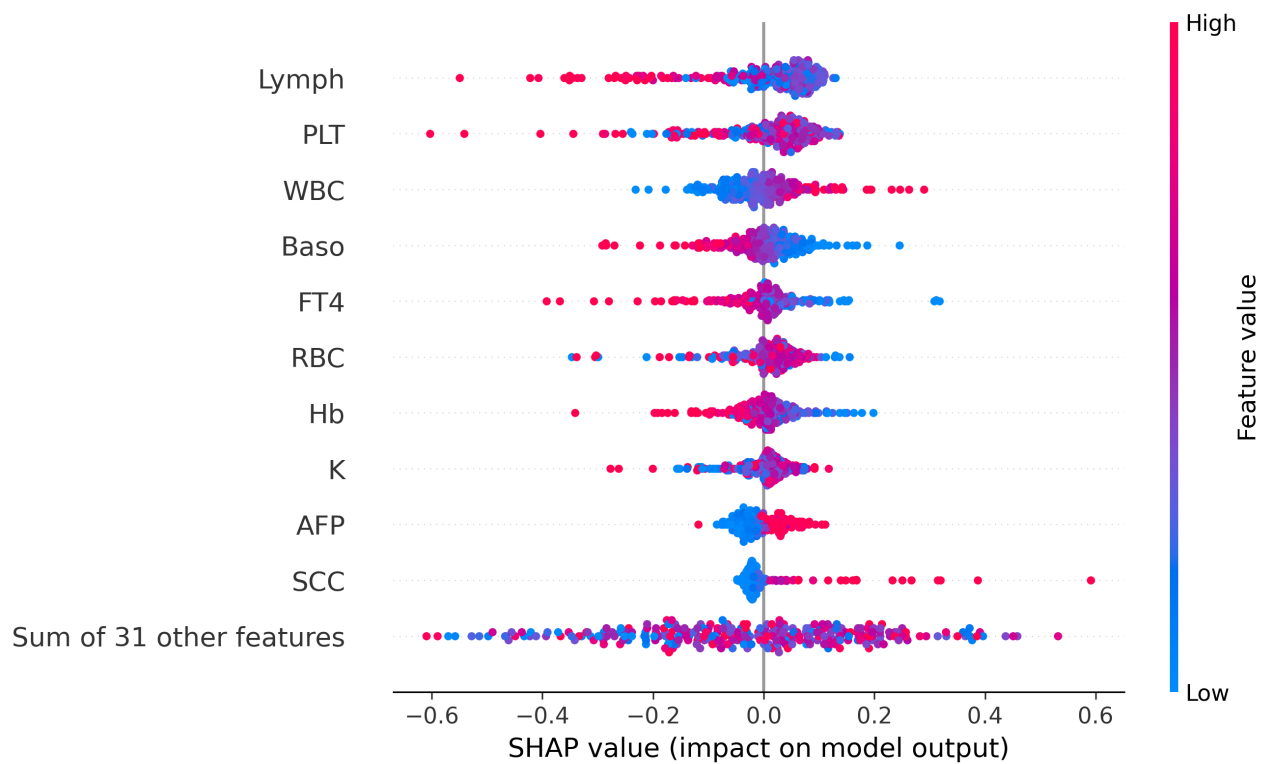


Figure 1: Shapley values of the top 10 features. The y-axis lists features ranked by their overall impact on the model prediction (mean absolute Shapley values). Each point corresponds to a single sample, with its horizontal position indicating the SHAP value (i.e., the contribution to the prediction). The color gradient represents the original feature value, where red indicates higher values and blue indicates lower values.

Clinical Indicators	Abbreviation
C-reactive protein	CRP
Total Bilirubin	TBIL
Total Cholesterol	TC
Estimated Glomerular Filtration Rate	eGFR
Albumin	ALB
High-Sensitivity Cardiac Troponin	hs-cTnI
Triglycerides	TG
Creatine Kinase-MB	CK-MB
Procalcitonin	PCT
Low-Density Lipoprotein Cholesterol	LDL-C
Creatinine	Cr
Uric Acid	UA
Alkaline Phosphatase	ALP
Alanine Aminotransferase	ALT
Gamma-Glutamyl Transferase	GGT
Aspartate Aminotransferase	AST
N-terminal pro-B-type Natriuretic Peptide	NT-proBNP
Alpha-Fetoprotein	AFP
Carbohydrate Antigen 15-3	CA15-3
Carcinoembryonic Antigen	CEA
Squamous Cell Carcinoma Antigen	SCC
Cytokeratin 19 Fragment	CYFRA21-1
Neuron-Specific Enolase	NSE
Carbohydrate Antigen 125	CA125
Carbohydrate Antigen 19-9	CA19-9
Progastrin-Releasing Peptide	proGRP
Free Prostate-Specific Antigen	FPSA
Total Prostate-Specific Antigen	TPSA
Anti-double-stranded DNA Antibody	anti-dsDNA
Cytoplasmic Anti-Neutrophil Cytoplasmic Antibody	cANCA
Anti-Nuclear Antibody	ANA
Anti-Smith Antibody	Anti-Sm
Anti-Cyclic Citrullinated Peptide Antibody	Anti-CCP
Anti-Centromere Antibody	CENP
Thrombin Time	TT
D-Dimer	D-Dimer
Activated Partial Thromboplastin Time	APTT
Red Blood Cell Count	RBC
Hemoglobin	Hb
White Blood Cell Count	WBC
Platelet Count	PLT
Eosinophil Percentage	EOS%
Basophil Percentage	BASO%
Lymphocyte Percentage	Lymph%
Free Thyroxine	FT4
Free Triiodothyronine	FT3
Thyrotropin Receptor Antibody	TRAb
Anti-Thyroglobulin Antibody	TGAb
Thyroid Peroxidase Antibody	TPO-Ab
Thyroid Stimulating Hormone	TSH
Treg (CD4+CD25briCD127dim) frequency in CD4+ T cells	CD4+CD25briCD127dim%
CD3+ T Lymphocytes Percentage	CD3+%
CD3+ CD4+ Helper/Inducer T Lymphocytes Percentage	CD3+CD4+%
CD19+ B Lymphocytes Percentage	CD19+%
CD3+HLA-DR- Resting T Lymphocytes Percentage	CD3+HLA%
CD14+ Monocytes Percentage	CD14+%
Lymphocytes (% of Nucleated Cells)	Lymphocytes
CD4+/CD8+ Helper/Suppressor T Lymphocyte Ratio (Th/Ts Ratio)	CD4+/CD8+
CD3+CD8+ Suppressor/Cytotoxic T Lymphocytes Percentage	CD3+CD8+%
CD3+HLA-DR+ Activated T Lymphocytes Percentage	CD3+HLA-DR+%
CD3+CD4+CD8+ Double Positive T Lymphocytes Percentage	CD3+CD4+CD8+%

Table 1: Clinical laboratory test indicators (Part 1)

Clinical Indicators	Abbreviation
CD3+CD16+CD56+ Natural Killer T Cells Percentage	CD3+CD16+CD56+%
CD20+ B Lymphocytes Percentage	CD20+%
CD3-CD16+CD56+ Natural Killer Cells Percentage	CD3-CD16+CD56+%
CD3-HLA-DR+ Activated B Lymphocytes Percentage	CD3-HLA-DR+%
Specific Gravity	SG
Squamous Epithelial Cells	Squamous
White Blood Cells(Urine)	WBC(U)
Red Blood Cells(Urine)	RBC(U)
Interleukin-1	IL-1
Interleukin-4	IL-4
Tumor Necrosis Factor-alpha	TNF- α
Interleukin-6	IL-6
Interleukin-10	IL-10
Tumor Necrosis Factor	TNF
Interleukin-8	IL-8
Interleukin-2	IL-2
Interleukin-5	IL-5
Potassium	K
Sodium	Na
Chloride	Cl
Complement C3	C3
Complement C4	C4
Patient Age	Age
Patient Hypertension	Hypertension
Patient Diabetes Mellitus	Diabetes Mellitus
Coronary Heart Disease	CHD

Table 1: Clinical laboratory test indicators (Part 2)